

Volt

Vibe2ship hackathon - coding ninjas X google for developers

Live Application link: <https://volt-839071485135.asia-southeast1.run.app>

Github repository link : <https://github.com/vaishbil/volt>

1. Problem statement selected: The last minute life saver

Students, professionals, and entrepreneurs frequently miss deadlines, assignments, meetings, and important commitments. Existing productivity tools rely on passive reminders that are easy to ignore and do little to help users actually complete their tasks.

2. Solution overview:

Volt is an AI-powered productivity companion designed to solve a deeply human problem: people don't miss deadlines because they forget, they miss them because existing tools do nothing more than send a passive notification that gets swiped away in seconds. Volt moves beyond reminders and into action.

Built on Google's Gemini AI and deployed via Google Cloud Run, Volt functions as a proactive productivity partner that thinks alongside the user, prioritizes their workload intelligently, plans their day around real commitments, and holds them accountable through habit tracking and immersive focus sessions all in one app.

The entire development, API integration, and deployment pipeline was powered through Google AI Studio.

The Core Problem Volt Solves:

Traditional productivity tools are passive. They remind you that something is due. They do not help you figure out what to do first, how long it will take, whether you can actually finish it today, or what happens if you don't start right now. Volt solves all four of these problems simultaneously.

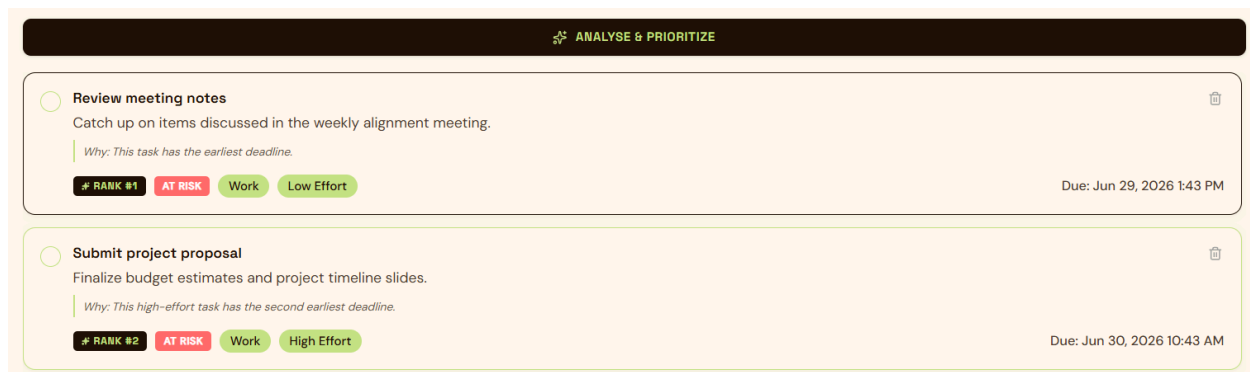
How Volt Works:

When a user opens Volt for the first time, they are greeted by a clean, minimal onboarding screen set against a dark espresso background with the Volt logo front and center. The screen asks for one thing only their name. For users who want to explore immediately without setting anything up, a "Try Demo Mode" option pre-loads sample tasks and habits so every feature of the app is instantly testable. By entering their name, the app becomes personal from that moment forward, greeting them by name every

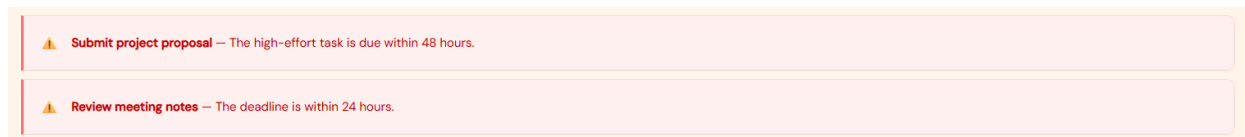
morning in the Streaks tab, addressing them in the Volt AI chat, and welcoming them back each time they return.

Task Manager :

In the Task Manager, users can add their tasks with a title. The description is optional, but adding one helps Gemini analyze and prioritize the task more accurately. Users are also required to add the deadline, effort level, and category. Tasks can be filtered based on their category.



When the user clicks "**Analyze & Prioritize**," Gemini analyzes all tasks simultaneously, weighs urgency against effort, assigns priority ranks, and explains exactly why each task is ranked where it is.



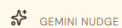
Simultaneously, a background risk detection system continuously monitors deadlines. Tasks that are due within the next 24 hours are flagged, and the system automatically surfaces alerts when something is at risk of being missed.

Users can also add tasks directly through **Volt AI Chat** using natural language. The AI automatically extracts the details and creates the task, which instantly appears in the **Task Manager** tab.

Habits and Goals:

Users can create habits and goals by defining a name, target duration, frequency (daily or weekly), and a goal period. for example, *"Morning Meditation, 15 mins, daily, for 30 days."* From that moment, Volt starts tracking the habit.

Users can log in every day and maintain streaks by marking a habit as completed with a single tap. The streak counter and fire badge update in real time. Each habit card can be expanded to view a progress map, where completed days are highlighted and future days remain empty. Users can also tap any past day to mark it as completed.



"Starting strong begins today. Dedicate a few minutes to your meditation and reading, and plan your fitness session for the week."

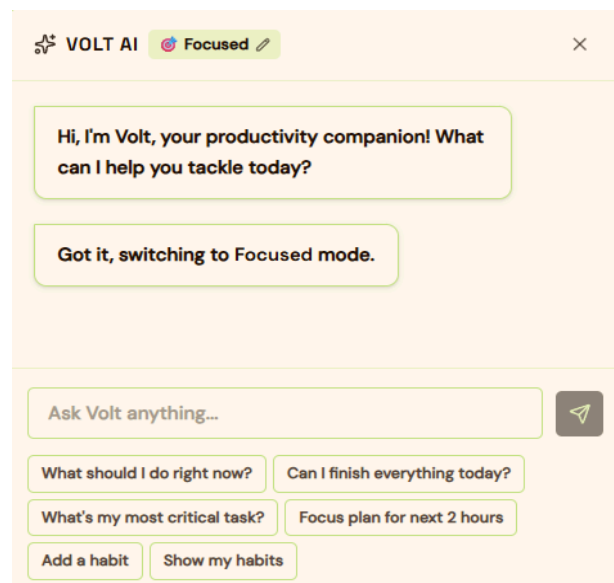
A Gemini-powered daily nudge appears at the top of the Streaks tab. Instead of showing a generic motivational quote, it generates a personalized message based on the user's actual habit data.

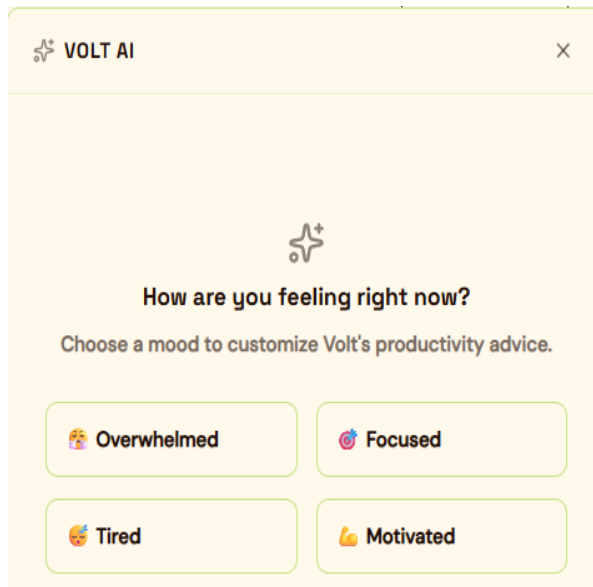
Users can also add habits and goals directly through **Volt AI Chat** using natural language. The AI automatically extracts the details and creates the habit or goal, which instantly appears in the **Habits & Goals** tab.

Volt AI (Chat Assistant):

Volt AI is not just a chatbot; it is a mood-aware productivity assistant that understands the user's tasks and habits and can take real actions on their behalf.

Before starting a conversation, users are asked to select their current mood:



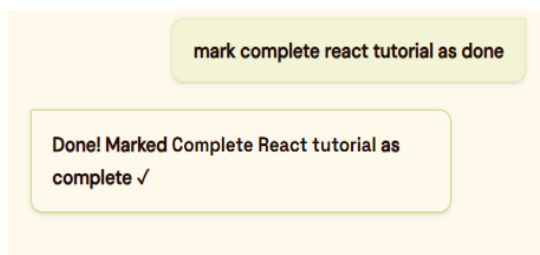


- **Overwhelmed** → Volt breaks tasks into smaller, manageable steps.
- **Focused** → Volt provides more aggressive task prioritization.
- **Tired** → Volt suggests lighter tasks first.
- **Motivated** → Volt matches the user's energy and pushes them harder.

The selected mood is displayed in the chat header and can be changed at any time during the conversation.

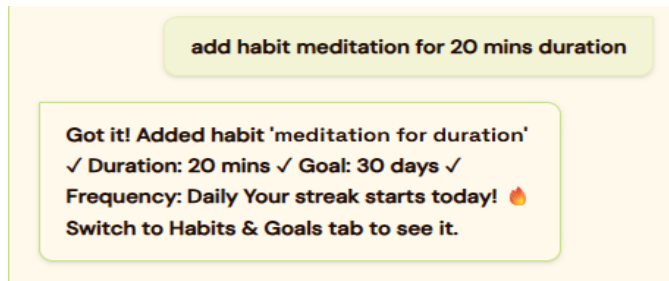
Task Commands:

- "add task finish report deadline tomorrow" → adds task instantly
- "mark react lecture as done" → marks task complete
- "delete office emails task" → removes task permanently



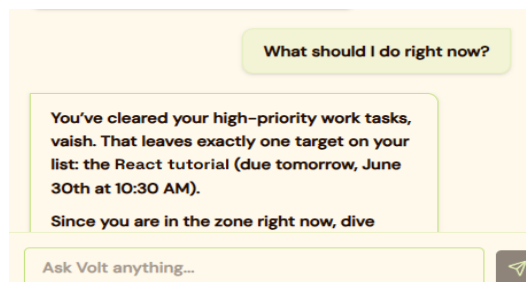
Habits and Goals Commands:

- "add habit morning yoga 20 mins for 30 days" → creates habit in Streaks tab
- "show my habits" → lists all current habits with streaks
- "add a habit" → Volt guides you through creating one

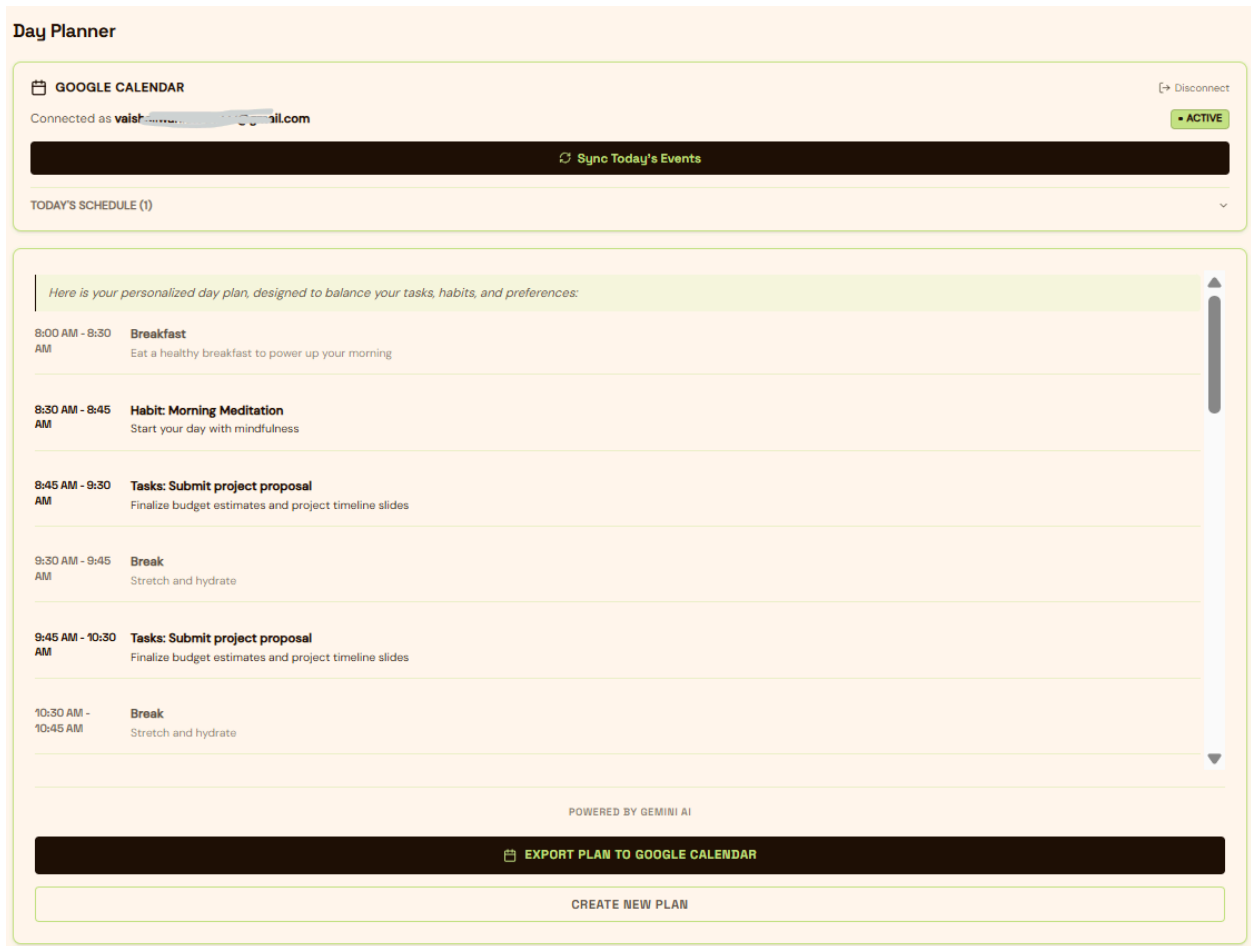


Ask Volt anything:

- "What should I do right now?"
- "Can I finish everything today?"
- "What's my most critical task?"
- "Focus plan for next 2 hours"

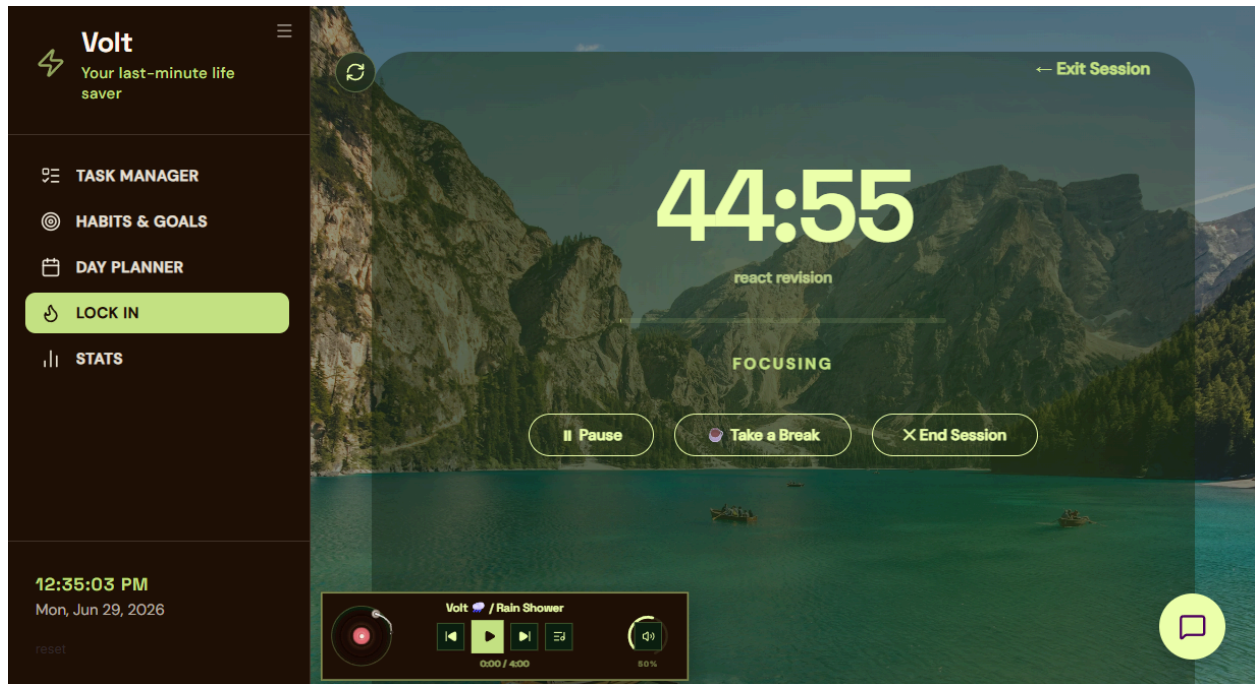


Day Planner :



Volt goes further than a standard task manager. It connects to the user's Google Calendar and fetches the day's meetings and commitments, then combines them with the user's tasks, habits, wake-up time, sleep time, and work style preferences. Gemini then generates a realistic, conflict-free, hour-by-hour schedule for the day, including meal breaks, energy levels, and focus blocks. The generated plan can be exported to Google Calendar with a single click, making it immediately actionable in the tools users already use.

Lock IN (focus Mode) :



Volt creates a distraction-free focus environment where users commit to a single task or habit for a set period. A Pomodoro-style timer counts down with structured break sessions, while ambient nature backgrounds and a vinyl-style music player provide a focused atmosphere. When the session ends, the completed task or habit is automatically marked as done across the app, with no manual updates required.

Users can also use **Lock-in Mode** for any activity that isn't already listed in their tasks or habits. The focus mode also includes multiple background options that users can switch between at any time.

Stats:

The **Stats** tab serves as a personal productivity dashboard, giving users a clear overview of their progress. It displays an overall **Volt Score** out of 100 based on task completion and habit consistency, along with a personalized motivational message.

Users can view task statistics, including active, completed, at-risk, and overdue tasks, as well as category-wise completion progress. Habit statistics are displayed through a visual progress chart and weekly points earned based on completed habits.

A weekly overview shows the number of tasks completed each day over the past seven days, and the "**Share My Progress**" feature allows users to generate and share a snapshot of their productivity.

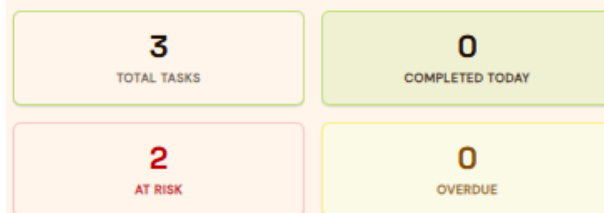
Hey vaish, here's your progress

Track your overall productivity metrics, streaks, and scores



You're crushing it!
YOUR OVERALL PRODUCTIVITY SCORE TODAY

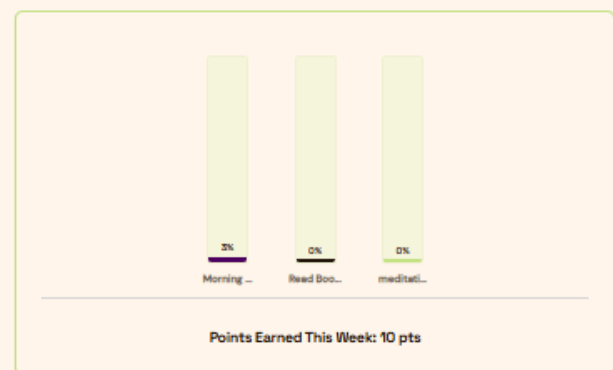
TASK STATS



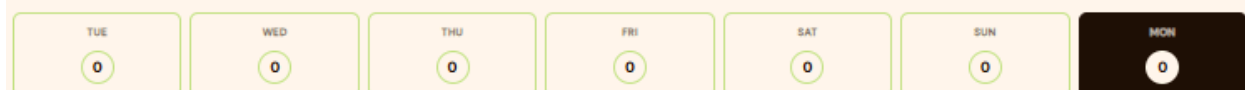
COMPLETION BY CATEGORY



HABIT STATS



WEEKLY OVERVIEW



🌟 SHARE MY PROGRESS

3. Key Features:

1. AI-Powered Task Prioritization

Gemini analyzes all tasks simultaneously, weighs urgency against effort and deadlines, assigns priority ranks, and explains why each task is ranked in that order.

2. Proactive Risk Detection

A background monitoring system continuously checks task deadlines and automatically flags tasks that are at risk of being missed.

3. Intelligent Day Planning with Google Calendar Sync

Volt syncs with Google Calendar and combines meetings, tasks, habits, wake time, sleep time, and work preferences to generate a realistic, conflict-free daily schedule that can be exported back to Google Calendar with a single click.

4. Mood-Aware AI Assistant

Before chatting, users select their current mood—Overwhelmed, Focused, Tired, or Motivated. Gemini adapts its responses and productivity suggestions based on the selected mood, which can be changed anytime during the conversation.

5. Habit Tracking with Streak Engine

Volt tracks daily and weekly habits, maintains streaks, displays a visual progress map, and generates personalized Gemini nudges based on the user's actual habit data.

6. Lock-In Focus Mode

A distraction-free focus environment featuring a Pomodoro timer, structured breaks, ambient backgrounds, nature sounds, and a vinyl-style music player. Completed tasks or habits are automatically marked as done when the session ends.

7. Agentic AI Chat

Volt AI understands natural language and can create tasks, add habits, mark items as complete, delete entries, and answer questions using the user's existing tasks and habits without requiring them to switch screens.

8. Unified Volt Score & Analytics

Volt calculates an overall productivity score based on task completion and habit consistency, while also providing task analytics, habit progress, weekly activity, and a shareable progress summary.

9. Built-in Ambient Soundscapes

The Lock-In mode includes browser-generated ambient sounds such as rain, campfire, stream, and drone, along with LoFi music to create a focused work environment without leaving the app.

4. Technologies Used

Core Framework & Language

- React v19 – Frontend UI library
- TypeScript v5.8 – Static typing

Build Tools

- Vite v6 – Fast frontend build tool
- ESBuild & tsx – TypeScript compilation

Styling & UI

- Tailwind CSS v4 – Utility-first responsive styling
- Lucide React – SVG icon library
- Space Grotesk & DM Sans – Google Fonts

Animations & Interactions

- Framer Motion – Animations and transitions
- dnd-kit – Drag-and-drop functionality
- Canvas Confetti – Celebration effects

AI & Backend

- Google Gemini API (@google/genai) – Powers all AI features
- Node.js & Express.js – Backend server and API routing
- React Markdown – Renders AI responses in chat

Audio

- Web Audio API – Procedural ambient sound generation
- HTML5 Audio – LoFi music playback

Data & Persistence

- localStorage – Offline-first client-side storage
- Firebase SDK – Authentication and Google Calendar OAuth

Deployment & Hosting

- Google Cloud Run – Serverless deployment
- Google AI Studio – AI development and deployment platform

Google AI Studio

- Volt was built and deployed using Google AI Studio as the primary development and deployment platform. The Gemini API was integrated and tested through Google AI Studio, and the application was containerized and deployed to Google Cloud Run directly via the AI Studio deployment workflow.

5. Google Technologies Utilized

1. Google Gemini AI (@google/genai)

- Powers task prioritization and risk detection.
- Generates AI day plans.
- Drives the mood-aware Volt AI Chat.
- Creates personalized habit nudges.
- Processes natural language commands.

2. Google Calendar API

- Fetches the user's meetings and events.
- Helps generate conflict-free daily schedules.
- Exports AI-generated plans back to Google Calendar.

3. Firebase Authentication

- Secure Google Sign-In with OAuth 2.0.
- Manages authentication for Google Calendar access.

4. Google AI Studio

- Used to build, test, and deploy the application.

- Integrated with Google Cloud Run.

5. Google Cloud Run

- Serverless hosting with automatic scaling.
- Hosts the deployed Volt application.

6. Google Fonts

- **Space Grotesk** for headings.
- **DM Sans** for body text and UI.

open source credits :

- React — MIT License
- Tailwind CSS — MIT License
- Framer Motion — MIT License
- dnd-kit — MIT License
- Lucide React — ISC License
- Canvas Confetti — MIT License
- React Markdown — MIT License
- SoundHelix (LoFi music) — Free for demo use
- Unsplash (ambient backgrounds) — Unsplash License